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Economic Commensalism:

Governing the Commons through Rational Incentives

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Abstract

Environmental degradation is often associated with a structural trade-off between increasing economic productivity and preserving the ecological systems that sustain production over time. Although economic theory has long considered nature through concepts such as land, natural resources, externalities, environmental costs, and common-pool resources, conventional production theory still tends to treat the environment primarily as a constraint, stock, or corrective problem rather than as an endogenous, dynamic, and strategically relevant component of production. This separation contributes to institutional arrangements in which preservation is frequently perceived by private agents as an external obligation, a compliance cost, or a restriction on productive activity.

This research proposes an alternative theoretical framework based on the concept of *economic commensalism*. The concept is used to describe an institutional and economic arrangement in which environmental preservation is not treated merely as a moral duty, legal constraint, or compensatory mechanism, but as a condition that can be incorporated into the agent's productive calculus. The central argument is that preservation may become individually rational when ecological variables are endogenized into production decisions and when incentives are spatially, intertemporally, and strategically calibrated. In this sense, the research moves beyond approaches that rely exclusively on Pigouvian taxation, Coasian bargaining, command-and-control regulation, or fragmented positive incentives. Instead, it investigates whether environmental preservation can alter the structural payoff of economic agents and become compatible with private productivity, profitability, and long-term welfare.

The thesis is structured around three complementary components. First, it develops a theoretical framework that revisits the relationship between production, preservation, extractivism, predation, and common-pool resource governance. Second, it proposes a formal model in which environmental variables, intertemporal decision-making, uncertainty, bounded rationality, and strategic interaction are incorporated into the analysis of agents' choices. Third, it evaluates the empirical feasibility of the proposed framework by applying it to the Brazilian state of Mato Grosso, a territory marked by agricultural expansion, deforestation pressures, and ecological heterogeneity across biomes.

The research therefore seeks to contribute to environmental and resource economics by reframing environmental preservation as a potentially endogenous component of economic rationality. Its main contribution is to examine whether locally grounded and incentive-compatible governance mechanisms can reduce the opposition between production and preservation, making ecological conservation not only socially desirable, but also strategically stable and economically rational for private agents.

1. Introduction

The relationship between economic activity and the natural environment has long been marked by a persistent tension between productive expansion and ecological preservation. Economic theory does not ignore nature. Land, natural resources, environmental costs, externalities, and common-pool resources have all been extensively discussed in different strands of the literature (PIGOU, 2010; HARDIN, 1968; OSTROM, 1990; DASGUPTA, 2001). The problem addressed in this thesis is more specific: in conventional production theory, the environment is rarely treated as an endogenous, dynamic, and strategically relevant component of production decisions. Instead, it often appears as a stock, a constraint, a source of externalities, or an object of corrective regulation.

This theoretical positioning has important implications. When the environment is treated mainly as an external condition of production, environmental preservation tends to be perceived by private agents as a legal obligation, a compliance cost, or a restriction on productive activity. In this setting, the economic agent is frequently placed before a trade-off: to preserve ecological assets in the interest of collective welfare or to expand productive use in the interest of private profitability. The central difficulty is not merely ethical or legal, but economic. If preservation does not alter the agent's expected payoff in a stable and predictable way, compliance will tend to depend on coercion, moral commitment, reputational pressure, or fragmented compensatory mechanisms.

This problem becomes particularly visible in the case of land use. Standard economic models usually define production as a function of land or natural resources, capital, and labor. Although ecological conditions are directly or indirectly embedded in these factors, the environment itself is often treated as exogenous to the productive process (JEHLE; RENY, 2011). Ecological functions such as soil fertility, water regulation, biodiversity, carbon sequestration, resilience, and regenerative capacity may affect productivity over time, but they are not always represented as variables that enter the agent's decision-making structure. As a result, environmental degradation is frequently addressed only after it occurs, through taxes, fines, restrictions, compensations, or restoration obligations.

The main institutional responses to environmental degradation reflect this logic. Pigouvian taxation seeks to internalize negative externalities by increasing the private cost of socially harmful activities (PIGOU, 2010). Property-rights approaches and Coasian bargaining emphasize the role of rights allocation and negotiation in correcting externalities. Command-and-control regulation imposes restrictions, prohibitions, and sanctions on environmentally harmful conduct. Common-pool resource theory shows that, under certain institutional conditions, communities may self-govern shared resources without necessarily relying only on privatization or centralized state control (OSTROM, 1990; OSTROM, 2002). Carbon markets, environmental certifications, payments for environmental services, and fiscal incentives attempt to create positive economic rewards for

conservation or reduced degradation.

These mechanisms have contributed significantly to environmental policy design. However, they also present limitations. Negative incentives, such as fines, embargoes, and legal restrictions, may deter illegal behavior, but they do not necessarily make preservation economically attractive. Positive incentives, such as carbon credits, certifications, subsidies, and payments for environmental services, may generate additional revenue, but they often remain fragmented, uncertain, or detached from the ecological unit affected by production. In many cases, environmental preservation remains an external condition attached to production rather than an endogenous component of the production process itself.

A second limitation concerns the spatial design of environmental compensation. Ecological degradation is territorially specific. A loss of native vegetation, biodiversity, hydrological function, or soil resilience occurs in a concrete ecological unit. Yet many compensation mechanisms operate through standardized, centralized, or globally tradable instruments. A producer may offset environmental impacts in a location different from the territory affected by production. This may satisfy an accounting or regulatory requirement, but it does not necessarily restore the ecological functions of the affected biome, watershed, ecosystem, or productive region. The problem, therefore, is not only whether compensation occurs, but where it occurs and whether it preserves the ecological conditions that sustain local production and long-term resilience.

A third limitation concerns time. Environmental degradation and preservation are intertemporal processes. The benefits of exploiting natural resources may be immediate, private, and measurable, whereas the benefits of preservation are often delayed, uncertain, collective, and spatially diffuse. This temporal asymmetry weakens the private incentive to preserve, especially when the agent faces credit constraints, competitive pressure, incomplete information, weak enforcement, or political uncertainty. As a result, even agents who understand the long-term importance of ecological preservation may still choose short-term expansion if the expected private payoff of degradation exceeds the expected private payoff of preservation.

These limitations suggest that the opposition between production and preservation may be less a natural incompatibility than a consequence of theoretical and institutional misalignment. If ecological variables remain external to production theory, and if environmental policy operates mainly as a corrective layer imposed after private decisions have already been made, preservation will tend to appear as a burden. Conversely, if ecological preservation is incorporated into the productive calculus of agents, it may become possible to design institutional arrangements in which preserving environmental assets becomes compatible with productivity, profitability, and long-term welfare.

This thesis develops this possibility through the concept of *economic commensalism*. The term is used as an analytical framework for describing economic arrangements

in which environmental preservation is not treated merely as a moral duty, legal constraint, or compensatory obligation, but as a condition that may enter the agent's rational decision-making process. The concept does not deny the relevance of regulation, enforcement, markets, or collective governance. Rather, it seeks to reorganize their role around a central question: under what conditions can environmental preservation become individually rational, strategically stable, and socially desirable?

The proposed framework differs from predatory and extractivist patterns of resource use. Predation refers to the destructive, illegal, or unsustainable depletion of natural resources, often disregarding property rights, ecological limits, and collective welfare. Extraction, in a neutral sense, refers to the use or removal of natural resources as part of productive activity. Extractivism, by contrast, is understood here as a productive and institutional regime in which resource use becomes structurally disconnected from ecological regeneration, future productivity, and the alignment between private and social payoffs. The distinction is important because not every productive use of nature is necessarily predatory; however, productive use becomes extractivist when it systematically undermines the ecological conditions upon which future production depends.

From this perspective, the central issue is not simply whether natural resources should be used or preserved. The relevant question is how institutions, incentives, and production theory can distinguish between productive use that maintains ecological functionality and productive use that erodes the basis of future production. Economic commensalism is proposed as a framework for investigating this distinction. It assumes that preservation may become economically rational when the benefits of maintaining ecological integrity are linked to the agent's expected returns, access to markets, credit conditions, production rights, fiscal treatment, reputational gains, technological monitoring, and long-term productivity.

The Brazilian state of Mato Grosso provides a particularly relevant empirical and conceptual context for this investigation. The state is one of Brazil's main agricultural frontiers and contains portions of three major biomes: the Amazon, the Cerrado, and the Pantanal. It is therefore located at the intersection of agricultural expansion, deforestation pressure, biodiversity conservation, carbon-related opportunities, and institutional disputes over land use. The ecological heterogeneity of Mato Grosso makes it especially useful for analyzing whether environmental governance should be designed according to broad political boundaries or according to ecological units such as biomes, watersheds, ecoregions, and transition zones.

Brazilian environmental law already reflects, at least partially, the importance of ecological differentiation. The Brazilian Forest Code establishes different legal requirements for native vegetation preservation depending on the region and biome in which rural properties are located (BRASIL, 2012). However, legal restrictions alone do not solve the incentive problem. If producers bear immediate opportunity costs from pre-

servation while receiving uncertain, delayed, or diffuse benefits, the law may generate resistance, circumvention, or dependence on costly enforcement. This reinforces the need to investigate institutional mechanisms capable of making preservation compatible with the agent's economic calculus.

The thesis therefore rests on four central premises. First, the environment can be represented as an endogenous cofactor of production, rather than merely as an externality or constraint. Second, many existing environmental instruments fail to alter the structural private payoff that drives degradation. Third, preservation becomes stable only when it is compatible with individual rationality and strategic interaction among agents. Fourth, such compatibility requires governance mechanisms that are locally anchored in ecological units while remaining connected to broader systems of verification, enforcement, and institutional credibility.

The general research problem can be stated as follows: can environmental preservation be theoretically reframed not merely as a constraint on production, but as an endogenous and incentive-compatible component of economic decision-making? More specifically, the thesis asks whether private productivity and profitability can be reconciled with socially desirable environmental preservation through institutional arrangements that are spatially grounded, intertemporally coherent, and strategically stable.

The main hypothesis is that, when ecological variables are incorporated into production theory and when incentives are designed to reflect spatial, intertemporal, and behavioral constraints, environmental preservation can become compatible with individual rationality rather than dependent solely on coercion, moral commitment, or fragmented compensation. In this sense, the thesis investigates whether economic commensalism can provide a theoretical basis for aligning private incentives with ecological preservation without necessarily reducing the long-term productive viability of economic agents.

The objective of the thesis is to develop the conceptual, formal, and empirical foundations of this framework. The theoretical objective is to clarify the limitations of approaches that treat environmental degradation mainly as an externality, a property-rights problem, or a regulatory failure. The formal objective is to model the interaction between production, preservation, uncertainty, bounded rationality, local ecological feedback, and strategic equilibrium. The empirical objective is to evaluate the feasibility of the proposed framework using Mato Grosso as a case in which agricultural productivity, deforestation, ecological heterogeneity, and environmental governance interact in a concrete institutional setting.

The thesis is organized into three main chapters. Chapter 1 develops the theoretical foundations of economic commensalism by discussing the relationship between production theory, environmental preservation, extractivism, predation, common-pool resources, and incentive-compatible governance. Chapter 2 formalizes the proposed framework through a mathematical model that incorporates environmental variables into production decisions,

with attention to intertemporal choice, uncertainty, learning, and strategic interaction. Chapter 3 provides an empirical application to Mato Grosso, examining whether the theoretical mechanisms proposed in the previous chapters are consistent with observed economic and environmental dynamics.

By reframing environmental preservation as a potentially endogenous component of economic rationality, this thesis seeks to contribute to environmental and resource economics, institutional economics, and public policy design. Its central contribution is not to reject existing instruments such as regulation, enforcement, carbon markets, payments for environmental services, or community governance, but to investigate how these instruments can be reorganized around a more precise theoretical problem: the need to transform preservation from an external burden into a strategically rational and locally grounded component of production.

2. Research Design

Building on the conceptual discussion presented in the Introduction, this section outlines the research design of the thesis. The study combines theoretical analysis, formal modeling, and empirical investigation to examine whether environmental preservation can be reframed not merely as an external restriction on production, but as an endogenous and incentive-compatible component of economic decision-making.

The research is guided by the concept of *economic commensalism*, understood here as a theoretical framework in which ecological preservation may become compatible with private productivity and profitability when incentives are spatially grounded, intertemporally coherent, and strategically stable. The central purpose is not to reject existing environmental instruments, such as regulation, enforcement, carbon markets, payments for environmental services, or community governance, but to investigate how these instruments can be reorganized around the structural payoff of economic agents.

The thesis is organized into three complementary chapters. Chapter 1 develops the theoretical foundations of economic commensalism, discussing the relationship between production theory, environmental preservation, extractivism, predation, common-pool resources, and incentive-compatible governance. Chapter 2 formalizes the proposed framework through a dynamic model that incorporates environmental variables, intertemporal choice, uncertainty, bounded rationality, learning, and strategic interaction. Chapter 3 evaluates the empirical feasibility of the framework through an application to the Brazilian state of Mato Grosso, a territory marked by agricultural expansion, deforestation pressures, and ecological heterogeneity across biomes.

Together, these three components seek to provide conceptual clarity, formal consistency, and empirical plausibility to the central claim of the thesis: environmental preservation may become economically rational when ecological variables are incorporated into production decisions and when institutional incentives are designed according to spatial, temporal, and behavioral constraints.

2.1. Research Problem

The research problem emerges from a specific limitation in conventional approaches to environmental economics and policy design. Although environmental degradation is widely recognized as a problem of externalities, common-pool resources, institutional failure, and intertemporal welfare, environmental preservation is still frequently treated as an external obligation imposed on private agents. In this setting, producers often face preservation as a legal constraint, compliance cost, reputational requirement, or compensatory burden, rather than as a component of their own productive strategy.

This thesis therefore asks whether the apparent conflict between private profitabi-

lity and socially desirable environmental preservation is inevitable, or whether it results, at least in part, from theoretical and institutional misalignment. If ecological variables are treated as exogenous to production, and if environmental policy operates mainly as a corrective layer imposed after private decisions have already been made, preservation will tend to appear as a burden. Conversely, if ecological preservation is incorporated into the productive calculus of agents, it may become possible to design institutional arrangements in which preserving environmental assets is compatible with productivity, profitability, and long-term welfare.

Accordingly, the central research question is:

> *Can environmental preservation be theoretically reframed as an endogenous and incentive-compatible component of economic decision-making, capable of reconciling private productivity and profitability with socially desirable ecological outcomes?*

This question guides the theoretical, formal, and empirical investigation developed throughout the thesis. An affirmative answer would indicate that environmental preservation can be designed as part of the agent's rational economic calculus, rather than depending exclusively on coercion, moral commitment, or fragmented compensation. A negative or partial answer would still be relevant, since it would clarify the limits of incentive-compatible environmental governance and identify the conditions under which preservation cannot be reconciled with private profitability without stronger forms of regulation or redistribution.

2.2. Hypotheses

The hypotheses of the thesis are structured around one general hypothesis and five specific hypotheses. The general hypothesis expresses the central proposition of the research. The specific hypotheses decompose this proposition into theoretical, behavioral, spatial, institutional, and strategic dimensions.

2.2.1. General Hypothesis

The general hypothesis is that:

H0. *It is possible to design an economic-institutional framework in which environmental preservation becomes compatible with private productivity and profitability when ecological variables are endogenized into production decisions and incentives are spatially, intertemporally, and strategically calibrated.*

H0 does not assume that preservation and profitability are automatically compatible. Rather, it states that compatibility may emerge under specific theoretical and institutional conditions. These conditions are examined through the specific hypotheses below.

2.2.2. Specific Hypotheses

H1. *Endogenizing environmental variables into the production function is necessary to correct the underrepresentation of ecological systems in production theory and to improve the design of environmental governance mechanisms.*

H1 addresses the theoretical foundation of the thesis. If the environment is treated only as an externality, constraint, or fixed stock, its effects on productivity, resilience, and intertemporal welfare tend to be represented only after degradation occurs. By incorporating ecological variables into production decisions, the thesis investigates whether environmental preservation can be modeled as part of the productive structure itself.

H2. *Economic decisions regarding environmental exploitation or preservation are intertemporal and shaped by bounded rationality, incomplete information, uncertainty, and adaptive learning.*

H2 defines the behavioral and temporal nature of the problem. Environmental decisions are not purely static choices made under perfect information. Agents face uncertain ecological feedback, price fluctuations, institutional instability, credit constraints, enforcement uncertainty, and imperfect knowledge about long-term consequences. This makes the preservation–exploitation dilemma dynamic and requires a framework capable of representing learning, expectations, and intertemporal trade-offs.

H3. *Environmental incentives are more likely to be effective when they are spatially and ecologically targeted to the units affected by production, such as biomes, watersheds, ecoregions, or transition zones.*

H3 addresses the spatial dimension of environmental governance. Ecological degradation is territorially specific, while many environmental compensation mechanisms are standardized, centralized, or globally tradable. The hypothesis is that incentives detached from the affected ecological unit may satisfy accounting or regulatory requirements without preserving the ecological functions that sustain local production and resilience.

H4. *Properly calibrated positive or mixed incentives can alter the structural payoff of economic agents, making preservation more attractive than degradation under specific institutional and economic conditions.*

H4 addresses the institutional mechanism of the thesis. The argument is not that positive incentives are always superior to negative incentives, nor that coercion is unnecessary. Rather, the hypothesis is that incentives linked to credit, certification, fiscal treatment, market access, carbon valuation, production rights, or payments for environmental services may reduce dependence on punitive enforcement when they make preservation privately advantageous.

H5. *Under ecological feedback, bounded rationality, and calibrated incentives, the interaction among agents may converge toward a stable strategic equilibrium in which environmental preservation becomes individually rational and collectively beneficial.*

H5 synthesizes the strategic dimension of the thesis. Preservation becomes stable only if it is compatible with the incentives faced by agents in interaction with one another. The relevant question is therefore not merely whether preservation is socially desirable, but whether it can become a rational strategy in a dynamic setting marked by uncertainty, local ecological constraints, institutional rules, and strategic behavior.

2.3. Objectives

The objectives of the thesis follow the same structure as the hypotheses. The general objective states the main ambition of the research, while the specific objectives identify the analytical steps required to pursue it.

2.3.1. General Objective

The general objective of the thesis is:

G0. *To develop a theoretical, formal, and empirical framework for investigating whether environmental preservation can be incorporated into economic decision-making as an endogenous and incentive-compatible component of production.*

This objective is pursued through the articulation of environmental economics, institutional economics, microeconomic theory, dynamic modeling, and empirical analysis applied to Mato Grosso.

2.3.2. Specific Objectives

G1. *To examine the theoretical implications of representing the environment as an endogenous cofactor of production rather than merely as an externality, constraint, or fixed stock.*

This objective corresponds to H1 and provides the conceptual foundation of the thesis. It seeks to clarify how environmental variables may affect productivity, resilience, opportunity costs, and welfare over time.

G2. *To model environmental decision-making under intertemporal constraints, uncertainty, bounded rationality, incomplete information, and adaptive learning.*

This objective corresponds to H2 and supports the formal modeling developed in Chapter 2. It aims to represent the exploitation–preservation dilemma as a dynamic process rather than a static choice.

G3. *To investigate how spatial and ecological scales affect the efficiency and effectiveness of environmental compensation and incentive mechanisms.*

This objective corresponds to H3 and examines whether environmental governance should be designed according to broad political boundaries or ecological units such as biomes, watersheds, ecoregions, and transition zones.

G4. *To assess whether positive or mixed incentives can alter the private payoff associated with preservation and reduce reliance on exclusively coercive environmental policy.*

This objective corresponds to H4 and evaluates the role of instruments such as carbon credits, payments for environmental services, environmental certification, fiscal incentives, subsidized credit, enforcement, and conditional production rights.

G5. *To analyze whether incentive-compatible preservation can generate stable strategic outcomes under ecological feedback and interaction among heterogeneous agents.*

This objective corresponds to H5 and examines whether preservation can become a rational strategy in equilibrium, rather than merely a socially desirable outcome imposed from outside the agent's decision structure.

2.4. Methodological Strategy

The thesis adopts a mixed research strategy composed of three interdependent methodological layers: theoretical-conceptual analysis, formal modeling, and empirical application. Each layer corresponds to one chapter of the thesis and contributes to a different dimension of the general hypothesis.

The first layer is theoretical and conceptual. It examines the foundations of environmental and resource economics, common-pool resource theory, institutional economics, and production theory in order to identify how the environment has been represented in

economic reasoning. This layer introduces the concept of economic commensalism and distinguishes it from predation, extraction, extractivism, and strict preservation. Its purpose is to clarify the conceptual problem that motivates the thesis: the treatment of environmental preservation as an external obligation rather than as a potentially endogenous component of production.

The second layer is formal and deductive. It develops a mathematical model of economic commensalism in which agents make intertemporal decisions under uncertainty, bounded rationality, ecological feedback, and strategic interaction. The model is expected to use dynamic optimization, elements of game theory, stochastic processes, Bayesian updating, Markovian transitions, and filtering mechanisms where appropriate. The purpose of this layer is to examine the internal coherence of the proposed framework and to identify the conditions under which a commensal equilibrium may exist.

The third layer is empirical and applied. It evaluates the plausibility of the theoretical and formal propositions using data from Mato Grosso. The empirical strategy will combine economic, environmental, geospatial, and institutional data to examine the relationship between land use, agricultural productivity, deforestation, ecological heterogeneity, environmental policy, and incentive mechanisms. The precise empirical specification will depend on data availability, policy relevance, and the feasibility of identifying credible counterfactuals.

This methodological structure allows the thesis to proceed from conceptual definition to formal analysis and then to empirical assessment. The theoretical chapter defines the problem and the framework. The mathematical chapter tests the internal consistency of the proposed mechanism. The empirical chapter evaluates whether the mechanism is compatible with observed economic and environmental dynamics in a concrete territory.

2.5. Chapter Structure

2.5.1. Chapter 1: Theoretical Framework

- **Preliminary title:** *Economic Commensalism: A Theoretical Framework to Endogenize Nature in Production Theory*
- **Purpose:** Chapter 1 develops the conceptual foundations of economic commensalism. It reviews the treatment of nature in production theory, environmental economics, common-pool resource theory, and institutional analysis. It also introduces a typology of economy–environment interactions, distinguishing predation, extraction, extractivism, strict preservation, and economic commensalism.
- **Methodological approach:** The chapter adopts a theoretical-analytical methodology based on literature review, conceptual reconstruction, and heuristic modeling.

It does not aim to provide empirical proof at this stage, but to establish the conceptual conditions under which environmental preservation can be interpreted as an endogenous and strategically relevant component of production decisions.

- **Expected contribution:** The expected contribution of Chapter 1 is to clarify the theoretical gap that motivates the thesis and to provide a coherent conceptual vocabulary for the formal and empirical chapters that follow.
- **Preliminary outline:**

  i. Production theory and the environment: land, natural resources, externalities, common-pool resources, and the exogenous treatment of ecological systems.

  ii. Environmental economics and institutional responses: Pigouvian taxation, Coasian bargaining, regulation, enforcement, carbon markets, payments for environmental services, certifications, and common-pool governance.

  iii. Typology of economy–environment interactions: predation, extraction, extractivism, strict preservation, and economic commensalism.

  iv. Endogenizing environmental variables: theoretical justification for treating ecological systems as cofactors of production and as determinants of intertemporal productivity.

  v. Economic commensalism: definition, assumptions, theoretical contribution, institutional implications, and limits.

2.5.2. Chapter 2: Formal Model

- **Preliminary title:** *Economic Commensalism: A Dynamic Production Model with Environmental Endogeneity*
- **Purpose:** Chapter 2 examines the internal consistency of the economic commensalism framework through formal modeling. It investigates whether environmental preservation can become individually rational and strategically stable when ecological variables are incorporated into the agent's intertemporal decision problem.
- **Methodological approach:** The chapter adopts a formal-deductive methodology. It is expected to combine dynamic programming, game theory, stochastic processes, Bayesian learning, Markovian dynamics, and filtering mechanisms where these tools are mathematically justified by the structure of the model. The model will identify the conditions under which a commensal equilibrium may exist and compare it with predatory or extractivist baselines.
- **Expected contribution:** The expected contribution of Chapter 2 is to provide a formal representation of the thesis mechanism and to clarify the mathematical

conditions required for preservation to be compatible with private rationality and strategic equilibrium.

- **Preliminary outline:**

  i. Basic structure of the model: agents, state variables, control variables, environmental stock or quality, production, payoff, and constraints.

  ii. Intertemporal decision problem: Bellman equation, value function, preservation choice, exploitation choice, and incentive regimes.

  iii. Uncertainty and learning: ecological shocks, incomplete information, Markovian transitions, Bayesian updating, and informational noise.

  iv. Strategic interaction: interaction among agents, local ecological feedback, governance rules, best responses, and equilibrium conditions.

  v. Existence and stability: continuity, compactness, convexity or concavity conditions, fixed point arguments, convergence, and comparative analysis of incentive regimes.

  vi. Computational simulation: simulation of positive incentives, negative incentives, mixed regimes, local compensation, global compensation, and commensal outcomes compared with extractivist baselines.

2.5.3. Chapter 3: Empirical Application

- **Preliminary title:** *Economic Efficiency and Environmental Resilience: Empirical Evidence of Commensal Strategies in Mato Grosso*

- **Purpose:** Chapter 3 evaluates the empirical plausibility of the economic commensalism framework using Mato Grosso as a case study. The chapter examines whether economic productivity, land use, deforestation, ecological heterogeneity, and environmental incentives interact in ways consistent with the theoretical and formal propositions developed in Chapters 1 and 2.

- **Methodological approach:** The chapter adopts an empirical strategy combining econometric analysis, geospatial analysis, and institutional interpretation. The precise methods will be selected according to data availability, treatment definition, policy relevance, and identification feasibility. Potential methods include panel data models, difference-in-differences, propensity score matching, synthetic control, spatial econometric models, and geographically weighted regression.

- **Expected contribution:** The expected contribution of Chapter 3 is to assess whether the mechanisms proposed by the thesis are empirically plausible in a territory marked by agricultural expansion, deforestation pressure, ecological heterogeneity, and relevant environmental governance instruments.

- **Preliminary outline:**

  i. Mato Grosso as empirical context: agricultural expansion, land use, deforestation, biome heterogeneity, institutional arrangements, and environmental policy.

  ii. Relevant programs and mechanisms: carbon markets, REDD+, REDD Early Movers, Environmental Reserve Quotas, payments for environmental services, the Produce, Conserve and Include Program, and other state or federal initiatives.

  iii. Data: agricultural productivity, land use, deforestation, vegetation cover, carbon-related indicators, environmental compliance, rural properties, credit, market access, and socioeconomic controls.

  iv. Empirical strategy: model selection, identification assumptions, spatial scale, temporal scale, treatment definition, control groups, and robustness checks.

  v. Expected empirical contribution: evaluation of whether locally grounded and incentive-compatible mechanisms are associated with better environmental and economic outcomes.

2.6. Scope and Limitations

The thesis is primarily theoretical and exploratory, with a formal and empirical component designed to test the plausibility of the proposed framework. It does not claim that economic commensalism is universally applicable, nor that environmental preservation will always be compatible with private profitability. The argument is conditional: compatibility depends on ecological characteristics, institutional capacity, monitoring technology, incentive design, enforcement credibility, market structure, and the intertemporal preferences of agents.

The empirical chapter is also subject to important limitations. Data availability may restrict the level of spatial disaggregation, the time period analyzed, and the possibility of identifying causal effects. Some mechanisms relevant to economic commensalism, such as expectations, perceived opportunity costs, informal governance, and strategic behavior, may not be directly observable in available datasets. For this reason, the empirical analysis will be interpreted as evidence of plausibility rather than definitive proof of the entire theoretical framework.

A further limitation concerns the normative implications of the thesis. The first three chapters adopt a primarily positive approach: they seek to define, model, and empirically assess the conditions under which preservation may become economically rational. A fourth chapter may be developed later to discuss legal, institutional, and policy appli-

cations of the framework, but such normative proposals will depend on the theoretical, formal, and empirical results obtained in the preceding chapters.

2.7. Expected Contribution

The expected contribution of this thesis is threefold. First, it contributes to environmental and resource economics by proposing a conceptual framework in which ecological preservation is analyzed as a potentially endogenous component of production decisions. Second, it contributes to institutional economics and public policy design by emphasizing the importance of incentive compatibility, local ecological specificity, and intertemporal coordination. Third, it contributes empirically by applying the framework to Mato Grosso, a case in which agricultural productivity, deforestation pressure, ecological heterogeneity, and environmental governance interact in a concrete and policy-relevant setting.

The thesis therefore seeks to advance the discussion on environmental governance by shifting the central question from whether private agents should preserve nature to whether institutions can be designed so that preservation becomes a rational and strategically stable component of economic behavior.

3. Significance of the Study

This section presents the significance of the study by clarifying its expected theoretical, methodological, empirical, and practical contributions. The thesis is exploratory in the sense that it proposes a new conceptual framework, but it is not merely speculative. Its contribution depends on three cumulative steps: first, showing that environmental preservation can be meaningfully represented as an endogenous component of production theory; second, formalizing the conditions under which such representation may alter the agent's intertemporal payoff; and third, evaluating whether these conditions are empirically plausible in a concrete territory marked by productive expansion and ecological pressure.

The significance of the thesis therefore lies in its attempt to move the environmental problem from the margin of economic decision-making to the core of the production process. Instead of treating ecological preservation only as a restriction, compensation mechanism, or moral imperative, the study investigates whether preservation can become part of the agent's rational strategy when incentives are properly designed and locally grounded.

3.1. Expected Results

The research expects to show that endogenizing environmental variables into production theory may improve the analytical representation of the relationship between productivity, profitability, and ecological preservation. More specifically, the thesis expects to demonstrate that the environment can be treated not only as a source of externalities or as a fixed stock of natural resources, but as a dynamic cofactor that affects production possibilities, intertemporal welfare, ecological resilience, and the strategic behavior of agents.

The expected theoretical result is the formulation of a coherent framework of *economic commensalism*. This framework should make it possible to distinguish between different economy–environment interaction regimes, such as predation, extraction, extractivism, strict preservation, and commensalism. The central expectation is that commensalism may describe a configuration in which productive activity generates private returns while maintaining, or at least not degrading beyond critical thresholds, the ecological conditions that sustain long-term production.

The expected formal result is the identification of conditions under which environmental preservation can become individually rational. These conditions include the presence of ecological feedback, intertemporal incentives, uncertainty, bounded rationality, learning, monitoring capacity, and institutional credibility. The formal model developed in Chapter 2 is expected to clarify whether and how preservation may become a stable

strategic choice, rather than merely a socially desirable outcome imposed on agents from outside their decision structure.

The expected empirical result is not to prove the universal validity of economic commensalism, but to evaluate its plausibility in the case of Mato Grosso. The empirical chapter will examine whether economic productivity, land use, deforestation, ecological heterogeneity, and environmental incentives interact in a way consistent with the theoretical mechanism proposed in the thesis. Evidence of compatibility between productivity and environmental preservation under certain institutional or territorial conditions would strengthen the framework. Evidence of incompatibility would also be relevant, since it would help identify the limits of incentive-compatible environmental governance.

3.2. Conceptual Preview

The conceptual starting point of the thesis is that the environment can be represented as an endogenous and dynamic component of production. In conventional models, output is often described as a function of capital and labor, sometimes with land or natural resources treated as an additional input. However, ecological systems are rarely modeled as variables that influence production through resilience, regeneration, water regulation, soil quality, biodiversity, carbon sequestration, and risk reduction.

To represent this idea in a simplified form, the thesis considers a modified production function in which output depends on capital K, labor L, and an environmental variable E:

$$Y = F(K, L, E).$$

In this formulation, E does not represent nature as a passive stock to be depleted. Rather, it represents the ecological conditions that affect production possibilities over time. These conditions may include soil fertility, hydrological stability, climate regulation, biodiversity, carbon storage, ecosystem resilience, and other biophysical characteristics relevant to production.

A basic commensalist configuration would require that environmental quality contributes positively to output within a relevant domain:

$$\frac{\partial Y}{\partial E} > 0.$$

This condition means that improvements in ecological quality, or the preservation of critical ecological functions, may increase productive capacity or reduce production risk. However, the model should not assume unlimited environmental productivity. It is reasonable to impose diminishing marginal returns:

$$\frac{\partial^2 Y}{\partial E^2} < 0.$$

This condition indicates that, although environmental quality matters for production, additional increments in E do not generate infinite or linear increases in output. The thesis also considers the possibility of complementarity between environmental quality and conventional inputs:

$$\frac{\partial^2 Y}{\partial K \partial E} > 0, \quad \frac{\partial^2 Y}{\partial L \partial E} > 0.$$

These inequalities express the hypothesis that the marginal productivity of capital and labor may increase when ecological conditions are preserved or improved. For example, better soil and water conditions may increase the productivity of agricultural machinery, fertilizers, labor, technology, and managerial effort. Conversely, environmental degradation may reduce the effectiveness of these inputs by increasing production risks, pest incidence, drought exposure, soil exhaustion, or restoration costs.

This simplified structure does not yet prove economic commensalism. It only establishes the analytical possibility that the environment can enter the production function as a variable whose preservation may affect private returns. The formal task of Chapter 2 is to determine the conditions under which this relationship is mathematically coherent and strategically stable. The empirical task of Chapter 3 is to assess whether the relationship is plausible in Mato Grosso.

The thesis also introduces the idea of a *commensal advantage*, understood as the net gain associated with a commensalist strategy relative to an extractivist or predatory baseline. In simplified terms, if $P_c(x)$ denotes production under a commensalist strategy and $P_e(x)$ denotes production under an extractivist strategy for a given level of environmental pressure x, the commensal advantage can be represented as:

$$\mathcal{C} = \int_{x'}^{x^*} [P_c(x) - P_e(x)]\, dx.$$

The interval $[x', x^*]$ represents the domain in which ecological integrity is still maintained within an acceptable threshold. The expression does not imply that commensalism always dominates extractivism. Rather, it defines a measurable theoretical object: the accumulated difference between two production trajectories under ecological constraints. If $\mathcal{C} > 0$, the commensalist strategy generates a higher cumulative return within the relevant ecological domain. If $\mathcal{C} \leq 0$, the framework must explain why commensalism fails to outperform the alternative strategy under the specified conditions.

This conceptual preview helps organize the thesis in three stages. Chapter 1 provides the theoretical vocabulary needed to define the regimes of interaction between production and the environment. Chapter 2 formalizes the conditions under which a commensal advantage may exist. Chapter 3 evaluates whether empirical evidence from Mato Grosso is consistent with the proposed mechanism.

3.3. Theoretical Significance

The theoretical significance of the thesis lies in its attempt to reposition environmental preservation within economic reasoning. Much of environmental economics has treated degradation as an externality, a common-pool resource problem, a property-rights issue, or a regulatory failure. These approaches remain essential, but they often preserve a conceptual separation between production and preservation. Production occurs first; environmental policy then corrects, restricts, taxes, compensates, or regulates its consequences.

Economic commensalism seeks to reduce this separation. It proposes that ecological preservation may be modeled as a component of the production decision itself. This does not mean that all preservation is profitable, nor that markets alone can solve environmental degradation. The theoretical claim is narrower and more conditional: under specific ecological, institutional, and incentive conditions, preservation may alter the expected payoff of agents in a way that makes it compatible with private rationality.

The thesis also contributes by refining the vocabulary used to describe economy–environment interactions. Predation, extraction, extractivism, strict preservation, and commensalism are not treated as biological categories mechanically transferred into economics. They are used as analytical categories to distinguish different institutional and productive regimes. This distinction is important because the thesis does not assume that all use of nature is predatory. The relevant question is when productive use preserves the ecological conditions of future production and when it undermines them.

The proposed framework may therefore contribute to environmental and resource economics, institutional economics, and ecological economics by offering a bridge between production theory, incentive design, spatial governance, and intertemporal environmental decision-making.

3.4. Methodological Significance

The methodological significance of the thesis lies in its integration of three levels of analysis. The first level is conceptual and theoretical, concerned with defining the categories and assumptions of economic commensalism. The second level is formal and deductive, concerned with modeling the conditions under which environmental preservation may become rational for agents operating under uncertainty and strategic interaction. The third level is empirical and applied, concerned with evaluating whether the proposed mechanism is plausible in Mato Grosso.

This structure is relevant because the thesis does not rely exclusively on normative argumentation. It begins with a conceptual proposition, translates it into a formal model, and then confronts it with empirical data. In this sense, the study seeks to avoid two

common weaknesses in environmental debates: purely moral claims that do not alter incentives, and purely technical instruments that lack a coherent theoretical foundation.

Methodologically, the thesis also emphasizes the importance of spatial scale. Environmental degradation and preservation occur in specific ecological units. Therefore, empirical strategies that ignore biome-level, watershed-level, or territorial heterogeneity may fail to capture the actual relationship between production and ecological resilience. By using Mato Grosso as a case study, the research seeks to examine whether environmental incentives should be evaluated not only by aggregate outcomes, but also by their territorial and ecological adequacy.

3.5. Empirical Significance

The empirical significance of the study derives from the choice of Mato Grosso as the main application. The state is one of Brazil's most important agricultural frontiers and contains portions of three major biomes: the Amazon, the Cerrado, and the Pantanal. This makes it a relevant case for analyzing the interaction between agricultural productivity, deforestation, land-use regulation, carbon-related opportunities, and ecological heterogeneity.

Mato Grosso is especially useful because it concentrates both the problem and the opportunity addressed by the thesis. On the one hand, agricultural expansion and land-use change generate environmental pressure. On the other hand, the state's productive capacity, territorial diversity, and environmental relevance create a setting in which incentive-compatible preservation mechanisms may have significant economic and ecological effects.

The empirical chapter is expected to assess whether areas with different ecological characteristics, land-use trajectories, policy exposure, or incentive structures exhibit different patterns of productivity, deforestation, environmental compliance, and resilience. Depending on data availability, the analysis may combine geospatial data, agricultural data, deforestation data, environmental registries, policy information, and socioeconomic controls.

The empirical contribution will be interpreted cautiously. The thesis does not claim that Mato Grosso alone can validate a general theory. Rather, the case is used to examine whether the mechanisms proposed by the theoretical and formal chapters are observable or plausible in a concrete setting.

3.6. Practical and Policy Significance

The practical significance of the study lies in its potential implications for environmental policy design. If preservation can be represented as part of the agent's productive calculus, then policy instruments should not be evaluated only by their coercive power or

compensatory value. They should also be evaluated by their ability to alter the structural payoff associated with preservation.

This perspective has implications for carbon markets, payments for environmental services, environmental certifications, rural credit, fiscal incentives, land-use regulation, and monitoring systems. Under a commensalist logic, these instruments should be designed to reward verifiable preservation linked to the ecological unit affected by production. This means that environmental compensation should not be treated merely as a generic accounting mechanism, but as a spatially grounded process connected to local ecological functions.

The framework also suggests that punitive instruments and positive incentives should not be treated as mutually exclusive. Enforcement remains necessary when agents have incentives to degrade or circumvent rules. However, enforcement alone may be insufficient if preservation continues to impose immediate private costs with uncertain or delayed benefits. A mixed institutional design may be more effective when sanctions prevent predatory behavior while positive incentives make preservation economically attractive.

In this sense, economic commensalism may inform the design of policies that combine local ecological monitoring, performance-based incentives, credible enforcement, and long-term economic rewards. The goal is not to replace environmental regulation with market mechanisms, but to integrate regulation, incentives, and ecological information into a coherent governance structure.

3.7. Comparative Contribution

The following table summarizes the proposed contribution of economic commensalism in comparison with other perspectives. The table is not intended to exhaust the literature, but to clarify how the thesis positions its framework.

Tabela 1 – Comparative perspectives on the environment in economic and ecological theories

Element	Classical and Neoclassical Economics	Institutional Economics	Ecology and Environmental Ethics	Green Economy	Economic Commensalism
Land and nature	Production factor, resource stock, or source of rent.	Resource governed by property rights, rules, norms, and transaction costs.	Ecological foundation of life, resilience, and systemic interdependence.	Natural capital to be valued, protected, and incorporated into economic planning.	Dynamic ecological cofactor with productive, strategic, and intertemporal relevance.
Environment	Frequently treated as an externality, constraint, or corrective problem.	Common-pool resource subject to institutional design and collective-action problems.	System with intrinsic value and ecological limits.	Complementary asset linked to sustainability and intergenerational welfare.	Endogenous component of production decisions and local ecological feedback.
Incentive logic	Prices, marginal productivity, utility, and profit maximization.	Rules, governance structures, enforcement, norms, and transaction costs.	Ethical responsibility, ecological limits, and precaution.	Regulatory and financial incentives for sustainability.	Spatially grounded and intertemporal incentives that alter the payoff of preservation.
Treatment of time	Intertemporal optimization, discounting, and capital accumulation.	Institutional evolution, path dependence, and rule adaptation.	Irreversibility, resilience, thresholds, and ecological regeneration.	Long-term planning, ecological risk, and intergenerational equity.	Dynamic strategic interaction under uncertainty, learning, and ecological feedback.
Unit of analysis	Individual agent, firm, consumer, or representative producer.	Communities, organizations, property regimes, and governance arrangements.	Ecosystems, species interactions, biomes, and ecological processes.	Territory, population, economy, and environmental footprint.	Agent interacting with other agents and with local ecological systems.
Normative orientation	Allocative efficiency and welfare maximization.	Institutional stability and reduction of governance failures.	Ecological integrity and limits to degradation.	Sustainable development and natural capital preservation.	Compatibility between private rationality, production, and ecological resilience.

This comparison shows that economic commensalism is not proposed as a substitute for existing theories, but as a framework that reorganizes some of their insights around a specific analytical problem: how to make preservation compatible with the agent's productive calculus under spatial, intertemporal, and strategic constraints.

3.8. Research Opportunities

The thesis may open several avenues for future research. First, it may contribute to the development of production models that incorporate environmental quality as a dynamic variable. Second, it may support further investigation into local carbon markets, payments for environmental services, environmental reserve mechanisms, and credit instruments tied to verifiable ecological performance. Third, it may motivate empirical research comparing different ecological units, such as biomes, watersheds, ecoregions, and transition zones, in order to evaluate whether local governance improves environmental and economic outcomes.

A further research opportunity concerns the relationship between ecological preservation and financial valuation. If preserved ecological assets can reduce production risk, improve long-term productivity, or generate verified environmental services, they may become relevant to rural credit, insurance, certification, and investment decisions. However, this possibility must be approached cautiously. The financialization of nature can generate speculation, greenwashing, and weak environmental additionality if it is not tied to credible local monitoring and ecological performance.

The commensalist framework therefore suggests a research agenda rather than a ready-made policy solution. Its future development will depend on formal refinement, empirical testing, institutional feasibility, and legal compatibility.

3.9. Limitations

The study has important limitations. The first is conceptual. Economic commensalism is an emerging framework, and its categories require careful definition to avoid becoming merely metaphorical. The thesis must therefore show that terms borrowed from ecology, such as commensalism and predation, have analytical usefulness in economics and are not only rhetorical analogies.

The second limitation is formal. Translating ecological dynamics such as resilience, thresholds, regeneration, and spatial interdependence into mathematical models is complex. The model developed in Chapter 2 may require simplifying assumptions, and these assumptions may limit the scope of its conclusions. The thesis will therefore distinguish between what is formally demonstrated, what is simulated, and what remains a theoretical conjecture.

The third limitation is empirical. Data availability may restrict the level of spatial disaggregation, the time period analyzed, and the possibility of identifying causal effects. Some relevant variables, such as expectations, informal governance, perceived opportunity costs, strategic behavior, and local ecological feedback, may be difficult to observe directly. For this reason, the empirical chapter will be interpreted as an assessment of plausibility rather than definitive proof of the full theoretical framework.

The fourth limitation is institutional. Even if the framework is theoretically coherent and empirically plausible, its implementation would depend on monitoring capacity, legal design, enforcement credibility, property-rights clarity, political feasibility, and market acceptance. These dimensions may be discussed as implications of the thesis, but they cannot be fully resolved within the scope of the first three chapters.

Recognizing these limitations is essential to preserve the scientific integrity of the research. The aim of the thesis is not to claim that economic commensalism solves all environmental conflicts, but to investigate whether it provides a coherent and useful framework for rethinking the relationship between production, incentives, and ecological preservation.

4. Final Remarks

This thesis addresses a central limitation in the way environmental problems are commonly represented in economic reasoning: the tendency to treat the environment as an external constraint, a source of externalities, or a corrective policy problem, rather than as an endogenous component of production decisions. The research does not claim that economic theory has ignored nature. Rather, it argues that the environment is often positioned outside the agent's productive calculus, which helps explain why preservation is frequently perceived as a legal obligation, compliance cost, or moral requirement instead of a rational economic strategy.

The thesis therefore proposes the concept of *economic commensalism* as a theoretical framework for investigating whether environmental preservation can become compatible with private productivity and profitability. The central intuition is that ecological preservation may become individually rational when environmental variables are incorporated into production theory and when incentives are designed according to spatial, intertemporal, and strategic constraints. In this sense, the research seeks to move beyond the opposition between economic development and environmental protection by examining the conditions under which preservation may alter the structural payoff of economic agents.

The first chapter develops the theoretical foundations of this framework. It discusses how production theory, environmental economics, common-pool resource theory, and institutional approaches have treated the relationship between economic activity and nature. It also introduces a typology of economy–environment interactions, distinguishing predation, extraction, extractivism, strict preservation, and economic commensalism. This conceptual step is necessary to clarify that not every productive use of nature is necessarily predatory, but that productive use becomes problematic when it undermines the ecological conditions that sustain future production.

The second chapter translates this theoretical framework into a formal model. Its purpose is to examine whether environmental preservation can be represented as an intertemporal and strategic choice under uncertainty, bounded rationality, ecological feedback, and institutional incentives. The model is expected to identify the conditions under which preservation may become individually rational and strategically stable. In this part of the thesis, the relevant question is not whether preservation is normatively desirable, but whether it can become a rational strategy for agents operating within a properly calibrated institutional environment.

The third chapter evaluates the empirical plausibility of the proposed framework using the Brazilian state of Mato Grosso as a case study. Mato Grosso is a relevant context because it combines high agricultural productivity, deforestation pressure, ecological heterogeneity, and the presence of different environmental governance instruments.

By examining the interaction between land use, productivity, deforestation, ecological conditions, and incentive mechanisms, the empirical chapter seeks to assess whether the theoretical propositions developed in the previous chapters are compatible with observed territorial dynamics.

The expected contribution of the thesis is threefold. Theoretically, it seeks to reposition environmental preservation within production theory by treating ecological systems as potential cofactors of production rather than merely as external constraints. Formally, it aims to identify the conditions under which preservation may become compatible with private rationality and strategic equilibrium. Empirically, it intends to evaluate whether this mechanism is plausible in a concrete territory where production and preservation pressures coexist.

The thesis also has potential implications for public policy and institutional design. If environmental preservation can be incorporated into the agent's productive calculus, then instruments such as carbon markets, payments for environmental services, environmental certifications, rural credit, fiscal incentives, monitoring systems, and enforcement mechanisms should be designed not only to compensate or punish agents, but to alter the expected payoff associated with preservation. This does not imply replacing regulation with market incentives. Rather, it suggests the need for mixed arrangements in which credible enforcement prevents predatory behavior while positive incentives make preservation economically attractive.

At the same time, the argument remains conditional. Economic commensalism is not presented as a universal solution to environmental degradation. Its feasibility depends on ecological characteristics, institutional capacity, monitoring technologies, property-rights clarity, enforcement credibility, market structures, and the intertemporal preferences of agents. The thesis therefore does not aim to prove that preservation will always be profitable, but to investigate whether, under specific conditions, it can become a rational, incentive-compatible, and strategically stable component of economic behavior.

In this sense, the broader purpose of the research is to contribute to a more precise understanding of the relationship between production, incentives, and ecological preservation. Rather than asking only whether economic agents should preserve the environment, the thesis asks whether institutions can be designed so that preservation becomes part of what rational agents have reason to do. If this proposition proves theoretically coherent, formally consistent, and empirically plausible, economic commensalism may offer a useful framework for rethinking environmental governance in territories where economic development and ecological resilience are deeply interdependent.

A. Timeline

1.1. Research Schedule and Timeline

According to the official academic calendar of the doctoral program, the student's **enrollment began on January 25, 2024**, which also marks the starting point for the time count toward graduation requirements. Thus, **the maximum allowed period for submitting the final dissertation is October 25, 2028**, with the **deadline for registering for the qualifying examination set for June 1, 2026**. Within this timeframe, the research plan is structured to ensure consistent progress and early completion of key milestones, allowing time for feedback, adjustments, and deepening of the theoretical and empirical components as necessary.

Period	Milestones and Planned Activities
2nd Semester 2025	* Start of research activities. * Final refinement of the research design and hypotheses. * Initial literature review and consolidation of the theoretical basis for Chapter 1. * Drafting of the introduction and early theoretical articulation.
Year 2026	* Completion of Chapter 1. * Development of formal theoretical and mathematical models (Chapter 2). * Preparation for the qualifying exam (by June 2026). * Initial design of data structure and econometric strategy.
Year 2027	* Empirical validation and econometric modeling (Chapter 3). * Data collection and processing. * Testing of the Beta Models. * Integration of theoretical and empirical findings.
Year 2028 (until Oct)	* Final editing, adjustments, formatting, and submission of the dissertation. * Preparation for defense. * Optional deepening of case studies or production of related academic papers.

This schedule was designed not only to meet the program's formal deadlines, but also to anticipate potential difficulties in modeling, data access, or empirical validation. The timeline includes deliberate margins to accommodate unforeseen circumstances and provides flexibility for refining the research, deepening promising analytical aspects, and exploring new hypotheses that may emerge throughout the doctoral process.

B. On the Use of AI

This document employed generative artificial intelligence (AI) tools, including machine learning-based language models, to support writing assistance, idea organization, textual coherence verification and mathematical and logical revision.

In this sense, it is important to emphasize that AI was used only as an auxiliary tool, with all the curatorial work, critical analysis, theoretical modeling, mathematical formalization, statistical validation and intellectual contributions conceived by the main author of the writing.

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